

WASP-2b

R-1145

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_200727 · created 2026-07-31T00:34:36+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459057.8568 +/- 0.0061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.047
Transit depth (Rp/Rs)^2	2.3 +/- 1.43 [%]
Orbital Inclination (inc)	85.06 +/- 2.86
Airmass coefficient 1 (a1)	0.601 +/- 0.043
Airmass coefficient 2 (a2)	0.44 +/- 0.31
Scatter in the residuals of the lightcurve fit is	7.12 %
Best Comparison Star	#3 - [261, 311]
Optimal Aperture	2.5
Optimal Annulus	4.76
Transit Duration (day)	0.065 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.152 ± 0.047 vs 0.1326 (deviation 0.41σ)
Duration fit vs published	0.065 d vs 0.0737 d (deviation 0.27σ)
Mid-time O–C	15.7 min
Depth SNR	3.2
Residual scatter	7.12 %

Light curve

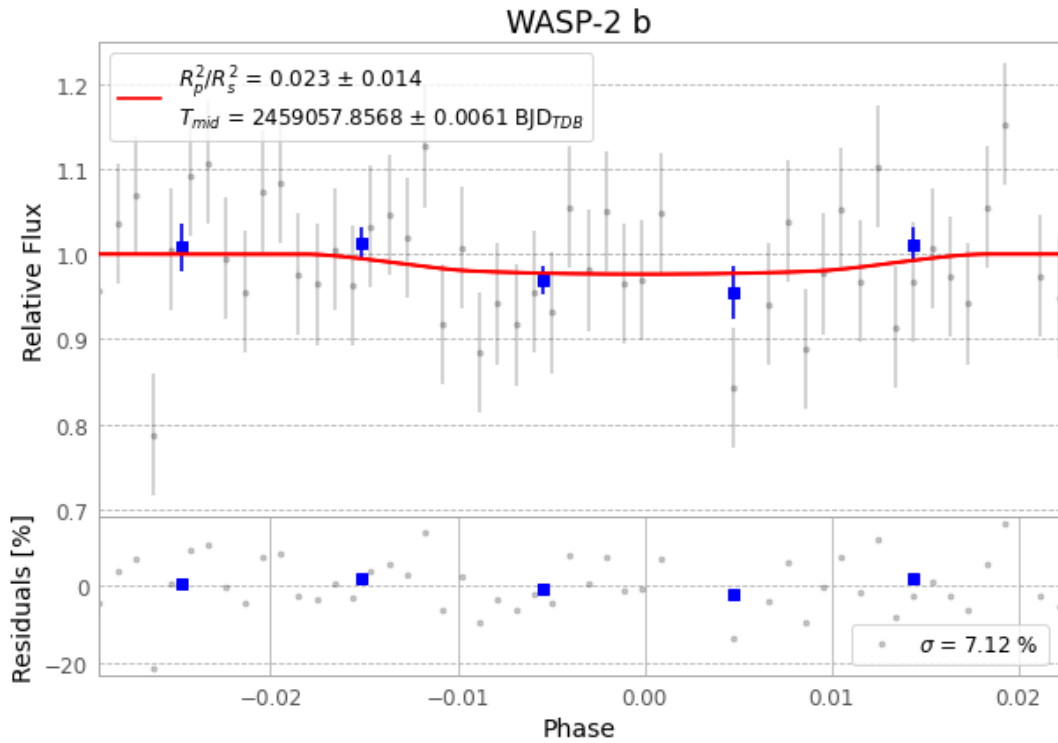


Figure 1 — FinalLightCurve_WASP-2 b_2020-07-26.png (SHA-256 7175f58a0c89cb92...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [219, 194], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8dc639413370db49...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

54 FITS frames (35 MB), WASP-2200727065412.FITS → WASP-2200727093312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-200727.log (ac1afc894ad4...), FinalLightCurve_WASP-2 b_2020-07-26.png (7175f58a0c89...), FinalLightCurve_WASP-2 b_2020-07-26.csv (dce9dd10aa9e...)

Compute resources

Wall time 160.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 00:34Z.